



Towards Better Design and Delivery of SE Team Project Courses

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ABSTRACT

Software Engineering (SE) team project courses allow students to explore the multifaceted challenges of software engineering. This doctoral thesis aims to conduct evidence-based research analyzing different aspects of SE team project courses. It will investigate the perspectives of students, teachers, and industry collaborators. The overarching goal is to enhance the design and delivery of these courses while aligning with industry needs.

CCS CONCEPTS

• **Social and professional topics** → **Software engineering education.**

KEYWORDS

Software Engineering Education, Project-Based Learning, Team Project Courses

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1 CONTEXT AND MOTIVATION

Software engineering (SE) is a human-intensive activity, and SE education requires an approach that goes beyond acquiring subject knowledge and technical skills [4]. SE students need to learn a diverse set of soft skills required for effective collaboration within teams [8, 11] and with industry clients [10].

Project-based learning (PBL) approaches are widely used in SE education [4, 12] to bridge the gap between theoretical knowledge and practical application [5]. Team project courses provide an opportunity to foster soft skills in SE students besides polishing their technical skills [5, 11–13]. There are several designs for project courses [12]; however, courses with industry clients provide a more realistic experience of a SE project [12]. Involving industry clients in SE team project courses (SETPCs) is not trivial [10]. These courses pose several challenges for students, teachers, and industry partners collaborating in these courses.

Students often underestimate complexities of teamwork and team dynamics until they engage in collaborative efforts [1, 12]. One of the factors contributing to teamwork complexity is diversity

within team. Diversity can imply heterogeneity across different dimensions, including age, sex, education, ethnicity, social background, and work experience [6]. Heterogeneous teams are more susceptible to conflicts than homogeneous ones [6]. Another challenge is teamwork in distributed settings (where not all team members are colocated) [4, 9]. In distributed settings, communication, coordination, and knowledge transfer within team, become more challenging [9]. Preparing students for distributed settings is crucial, as they are becoming increasingly prevalent in the software industry [9]. SETPCs also present challenges for teachers, requiring them to select effective methods for assessing soft skills and individual performance in a team [13]. In addition, these courses are inherently resource intensive, requiring continuous feedback and mentorship [3, 13], making them difficult to scale. Involving industrial clients in SETPCs [13] and ensuring their relevance to the industry is another challenge [4]. The industrial relevance of these courses is crucial as these help prepare students for their future careers [4, 13]. Cico et al. [4] argue that curricula for SE project courses should be updated according to trends in the software industry.

2 LITERATURE REVIEW

This section presents a brief account of the literature on two aspects relevant to this thesis, i.e., collaboration with industry clients and soft skills in SETPCs.

In a recent systematic literature review (SLR), Tenhunen et al. [13] have observed that a large number of project courses did not have an industrial client [13]. Similarly, in their mapping study, Cico et al. [4] observed a lack of industrial stakeholders' participation in SE project courses. Among 126 mapped studies, only 26 reported participation of industry stakeholders as mentors, clients, or customers. The authors emphasized the need to enhance participation of industry stakeholders and consider industry needs in design of SE project courses [4]. Paasivaara et al. [10] investigated the perspective of industrial collaborators for SETPCs. The authors have suggested a lack of existing evidence on how this collaboration should take place in practice [10]. Similarly, Ståhl et al. [12] have reported a lack of literature on needs and motivations of industry collaborators for these courses.

In SETPCs, a fundamental consideration is to create opportunities for fostering soft skills. Several studies prioritize the development of soft skills as a desired learning outcome in these courses [1, 7, 8, 11, 12]. A few studies present the need to prepare students for diverse [6] and distributed environments [4, 9] through SETPCs.

3 PROBLEM STATEMENT

The involvement of an industrial client in SETPCs can complicate the dynamics of these courses across all facets, necessitating all stakeholders to have excellent soft skills. From students' viewpoint,

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it presents minimal control over project scope, choice of tools and technologies, and working environment. It also complicates client communication and project management. Working with an industrial client lets the students experience the real external pressure [4]. From teachers' viewpoint, involvement of an industrial client implies a complicated course setup, less predictable course outcomes, and the stress of finding committed industry collaborators [13]. Despite these challenges, involving industrial clients is recommended for SETPCs [13], as this can prepare students for real-world scenarios within software industry. Based on the literature review presented in Section 2, it can be inferred that collaboration with industry clients in SETPCs is an under-researched area [10, 13]. There is a need to investigate further how the involvement of an industrial client influences the design and delivery of SETPCs.

4 RESEARCH GOALS

SETPCs are crucial to computing education as they allow students to practice not only technical skills but also soft skills. The overarching goal of our research is to enhance the design and delivery of SETPCs involving industry clients. This corresponds to three research objectives: 1) to identify the problems faced by the students, teachers, and industry collaborators in SETPCs, 2) to explore how the identified problems have been addressed in research and practice, and 3) to evaluate the identified solutions through implementation in our SETPCs. See Figure 1 for details.

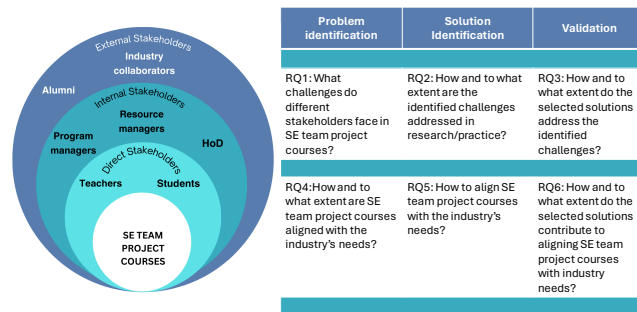


Figure 1: Research Overview

5 RESEARCH METHODS

The research will employ a combination of qualitative and quantitative methods. For **RQ1**, we have used thematic analysis of students' experience reports from SETPCs to identify their challenges. To explore challenges faced by teachers, we plan to conduct workshops. Next, we plan to conduct SLRs to identify solutions in existing literature (**RQ2**). We will conduct surveys and workshops to investigate the state of practice (**RQ2**, **RQ4**, **RQ5**). Finally, we plan to conduct case studies to validate identified solutions within the context of selected courses (**RQ3**, **RQ6**).

6 CURRENT & EXPECTED CONTRIBUTIONS

Our first contribution corresponds to **RQ1**. We have conducted a study to explore students' perspectives on challenges in SETPCs with industrial clients. As per our findings, most of these challenges relate to soft skills. The most reported challenge is working in a team. Factors contributing to this challenge include different

ethnicities and heterogeneity in the skill set of team members. Working in a distributed setting and working on a project with an industrial client are the next most reported challenges. Based on these findings, we plan to contribute further by exploring 1) how and to what extent SETPCs prepare students to work in diverse teams and what the best practices are for managing diversity in student teams, 2) how and to what extent our SETPCs prepare students to work in distributed work environments. Some of our colleagues already have competence in this area [2]. Building upon their work, we aim to identify best practices facilitating students in this setting, 3) exploring teachers' challenges in SETPCs to highlight potential improvement areas in these courses (contributions 1-3 map to **RQ2**), 4) Investigating how and to what extent SETPCs align with industry needs. Findings from this investigation would lead to improving alignment between SETPCs and industry needs (**RQ4** and **RQ5**), and finally, 5) evaluation of identified solutions through implementation in selected SETPCs, (**RQ3** and **RQ6**). This would be a significant contribution as the existing literature lacks evaluation research in SE project courses [4].

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